I. Introduction to Signals and Systems

1 Goal of Signals & Systems

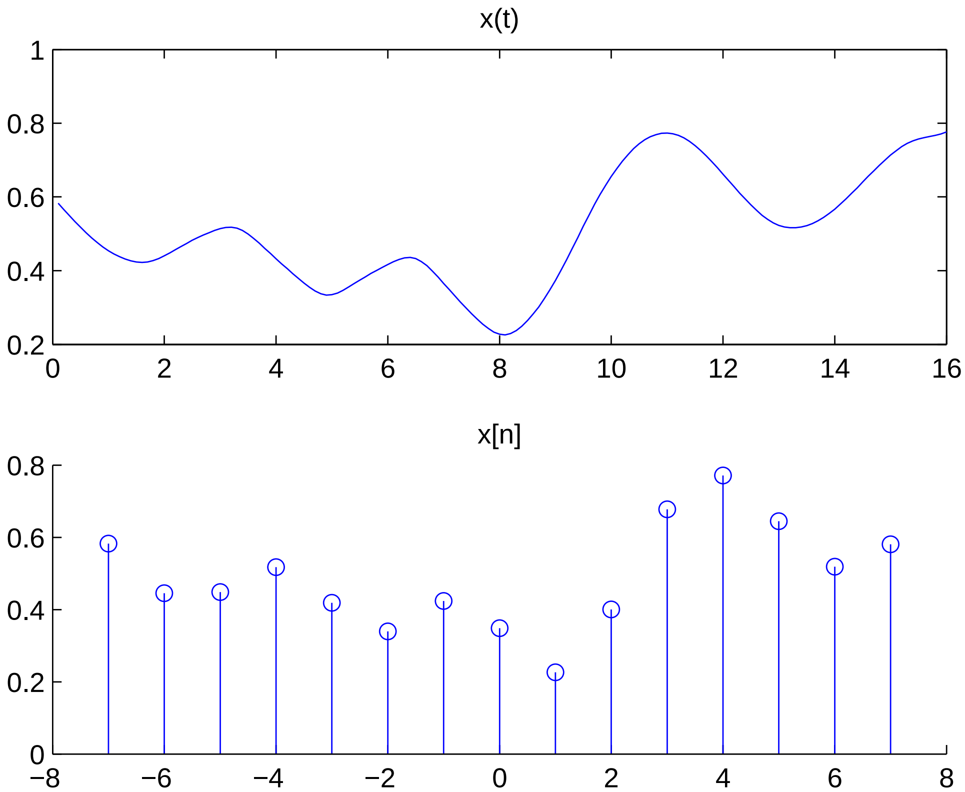
to develop mathematical models/techniques for continuous-/discrete-time signal and system analysis/synthesis(design).

- **Signals(functions)** are variables(time/space) that carry information,
  Ex. $f(t)$ speech(voltage/current picked up from microphone, a computer file xxx**.wav**, stock market index, ECG(ElectroCardioGram); $f(x,y)/\underline{f}(x,y)$ image xxx**.gif** or xxx**.jpg**; $f(t,x,y)$ video xxx**.mpg**.

- **Systems(mapping)** process input signals to produce output signals. They can be hardwares(DSP processors/circuits/automobile, economic system) or softwares(programs).

2 Classifications of signals

1. Digital/analog: the amplitude is discrete(quantized) or not.
2. Continuous-/Discrete-time: (MATLAB commands: `plot, stem`)
   - Continuous-time signal: $x(t)$, $-\infty < t < \infty$, where the time-variable t is continuous.
   - Discrete-time signal: $x[n] \equiv x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, where T represents the sampling time. Note that $x[3\frac{1}{2}]$ is NOT defined. Many discrete-time systems are computer programs.



3. Deterministic/Random:

- Deterministic signals:
  - periodic(Fourier series)

    $$x(t) = x(t+T), \quad \forall t$$

    minimum T is the fundamental period.
  - aperiodic(Fourier transform)
- Random signals(such as noise): probability density function, mean, autocorrelation, power spectral density.

4. Energy/Power signals:

- Size of a signal $x(t)$:
  - **energy**:

    $$E \equiv \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

    if $x(t)$ is the voltage applied across a dummy load of one-Ω resistor, then $x^2(t)$ is the power assumption (Watts). Integration of power over time is the total energy of the signal, namely, E indicates the energy that can be extracted from the signal.
    * Continuous-time: T
      Total energy over the time interval $t_1 \leq t \leq t_2$ is $\int_{t_1}^{t_2} |x(t)|^2 dt$.
    * Discrete-time: total energy over the time interval $n_1 \leq n \leq n_2$ is $\sum_{n=n_1}^{n_2} |x[n]|^2$.
  - **average power**:

    $$P \equiv \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

    * If $x(t)$ is periodic, then its average power becomes

      $$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

    * The rms(root-mean-square) value of $x(t)$ is $\sqrt{P}$.
- Definitions:
  - $x(t)$ is an energy signal if it has a finite energy $E_x < \infty$ and its power $P_x = 0$.
  - $x(t)$ is a power signal if it has a finite power $P_x < \infty$, and energy $E_x = \infty$.
- (Lathi Ex 1.2, p 72)
  - Power of a single real sinusoid $x(t) = C\cos(\omega_0 t + \theta)$ is $P_x = \frac{C^2}{2}$, where C is a positive amplitude.
  - Power of a single exponential $y(t) = De^{j\omega_0 t}$ is $P_y = |D|^2$, where D is a complex number comprised of amplitude and phase.
  - Power of sum of **orthogonal** complex exponentials $z(t) = \sum_i D_i e^{j\omega_i t}$ with distinct frequencies ω_i's is equal to the sum of individual powers $P_z = \sum_i P_i = \sum_i |D_i|^2$.
  - (Q) Suppose $z(t) = x(t) + y(t)$. Under what condition will the power of $z(t)$ be equal to the sum of those of $x(t)$ and $y(t)$, i.e., $P_z = P_x + P_y$?
- Examples:
  (a) $x(t) = e^{-3t}u(t)$ is an energy signal with energy 1/6.
  (b) $x(t) = Ae^{j2\pi\alpha t}$ is a power signal with power A^2.
  (c) A rectangular pulse $A\sqcap(t/\tau)$ of amplitude A and duration τ has energy $A^2\tau$.
  (d) $A\sqcap(t/\tau)\cdot\cos(\omega_0 t + \theta)$ has energy $A^2\tau/2$.
  (e) A rectangular pulse train $\sum_{n:-\infty}^{\infty} \sqcap(\frac{t-nT_0}{\tau})$, $T_0 > \tau$, is a power signal with $P = A^2\tau/T_0$

(f) A ramp signal $f(t) = t$ and an everlasting exponential $g(t) = e^{-at}$ are neither energy nor power signals.

(g) Is $x(t) = 10$ a power or energy signal?

(h) What is the energy of a sinc function $sinc(x) = \frac{\sin \pi x}{\pi x}$? (Parseval's thm in Chp 7's Fourier transform can solve it easily.)

3 Basic operations on signals

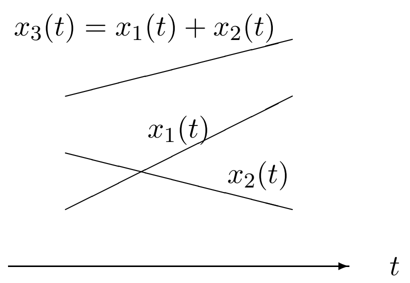
1. Amplitude:

   (a) amplitude-scale: $x(t) \Longrightarrow ax(t)$, where $a > 0$,
   - $a > 1$: amplify(gain > 1)
   - $0 < a < 1$: attenuate(gain < 1)

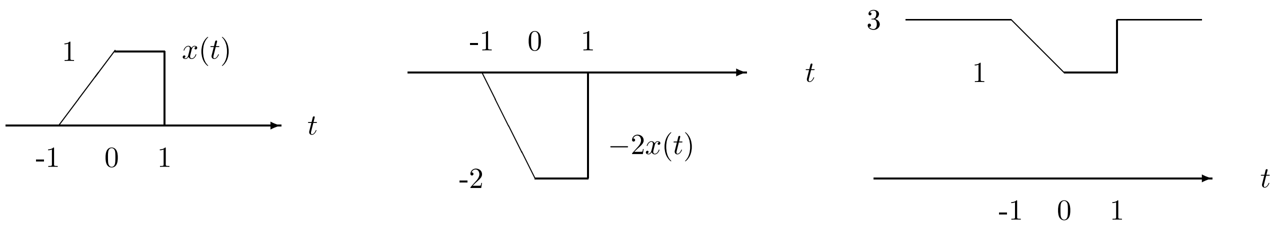
   (b) negation: $x(t) \Longrightarrow -x(t)$, change of polarity

   (c) level-shift: $x(t) \Longrightarrow x(t) + c$
   - $c > 0$: shift up by c
   - $c < 0$: shift down by c

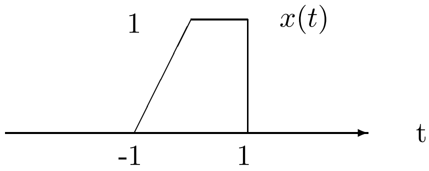
   (d) Note that sum of two lines is itself a line, too.



[Ex] Given $x(t)$, plot $y(t) = -2x(t) + 3$.
("multiply/divide first, then add/subtract")

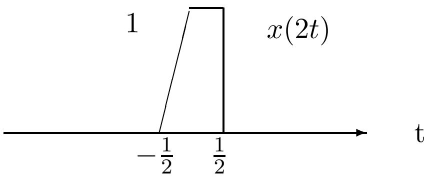


2. change of **Time**-variable: Assume that $x(t)$ looks like

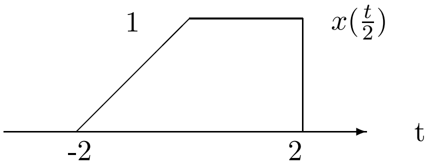


(a) time-scale: $x(t) \Longrightarrow x(\alpha t)$, where $\alpha > 0$

- $\alpha > 1$, **time-compression**:

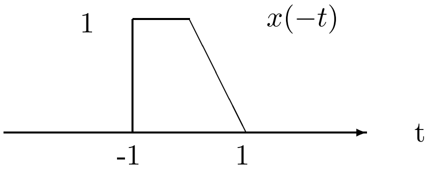


- $\alpha < 1$, **time-expansion**:



This may seem to hurt your instinct. For justification, Let $y(t) = x(2t)$. Choose a few t's and plot $y(t)$, we have $y(0) = x(0)$(still centered around the origin), $y(\frac{1}{2}) = x(2\frac{1}{2}) = x(1)$, and $y(-\frac{1}{2}) = x(-2\frac{1}{2}) = x(-1)$.

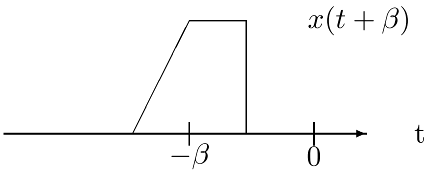
(b) time-reversal(inversion): $x(t) \Longrightarrow x(-t)$



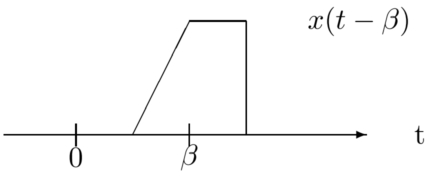
The right-hand-side is mirrored about the origin, and vice versa.

(c) time-shift: $x(t) \Longrightarrow x(t + \beta)$

- $\beta > 0$, left-shifted(advanced)



- $\beta < 0$, right-shifted(delayed):



Again, this seems to be unbelievable. A mindful reader will not hesitate to play the same game: Let $y(t) = x(t - 3)$. $y(3) = x(3 - 3) = x(0)$, $y(0) = x(0 - 3) = x(-3)$, *etc.* From which, she/he convinces herself/himself.

(d) Combined operation: $\boxed{x(t) \Longrightarrow x(\alpha t + \beta)}$,

[method 1]:

- Let $y(t) = x(t + \beta)$, *i.e.*, advance/delay $x(t)$ by β to obtain $y(t)$.
- Let $z(t) = y(\alpha t)$, *i.e.*, compress/expand $y(t)$ by α.

As a check, $z(t) = y(\alpha t) = x(\alpha t + \beta)$. The above procedure can change its order. BUT, be careful!

[method 2]:

- Let $v(t) = x(\alpha t)$, *i.e.*, compress/expand $y(t)$ by α.
- Let $\underline{w(t) = v(t + \frac{\beta}{\alpha})}$, INSTEAD OF $w'(t) = v(t + \beta)$.

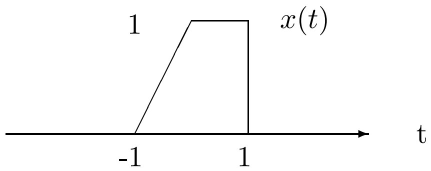
As a check, $w(t) = v(t + \frac{\beta}{\alpha}) = x(\alpha[t + \frac{\beta}{\alpha}]) = x(\alpha t + \beta) = z(t)$, while $w'(t) = v(t + \beta) = x(\alpha[t + \beta]) = x(\alpha t + \alpha\beta) \neq z(t)$!
Note:

- if you have difficulty in combining $w(t)$ and $v(t)$, you can use more dummy variables to avoid confusion:
$$v(s) = x(\alpha s), \quad w(t) = v(t + \frac{\beta}{\alpha})$$
so that
$$w(t) = v(s)\big|_{s=t+\frac{\beta}{\alpha}} = x(\alpha[t + \frac{\beta}{\alpha}]) = x(\alpha t + \beta)$$
- [Method 1] is preferred due to its algebraic simplicity.

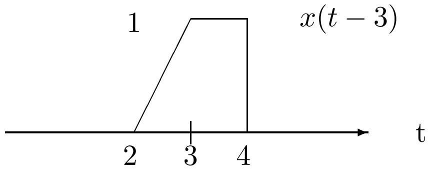
[Ex] Given $x(t)$, plot $y(t) = x(2t - 3)$, and $z(t) = x(-2t + 3)$.
("subtract/add first, then divide/multiply", a rule which is exactly opposite to the one used in the amplitude operations.)

It is strongly recommended to check $y(t)/z(t)$ at some t's where $x(t)$ changes abruptly(and $t = 0$).

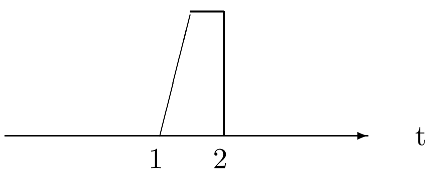
- Given $x(t)$



- Right shift by 3, $x(t - 3)$



- Compress by 2, $x(2t - 3)$

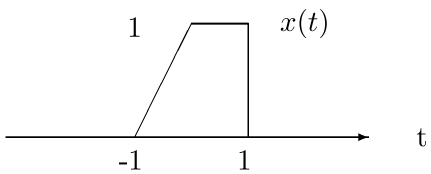


- (Check)

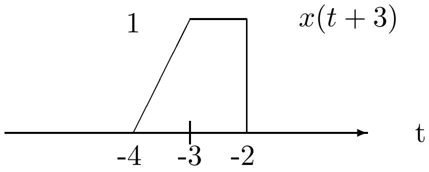
$$\begin{array}{lll} \text{let } x(2t-3) = x(0), & \text{we have } t = 3/2 & \text{indeed } y(3/2) = x(2t-3) = x(0) \\ \text{let } x(2t-3) = x(-1), & \text{we have } t = 1 & \text{indeed } y(1) = x(2t-3) = x(-1) \\ \text{let } x(2t-3) = x(2), & \text{we have } t = 5/2 & \text{indeed } y(5/2) = x(2t-3) = x(2) \end{array}$$

Similarly,

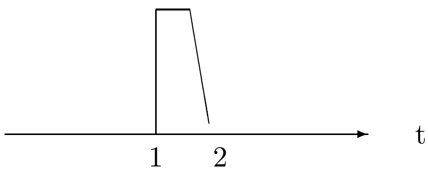
- Given $x(t)$



- Left shift by 3, $x(t+3)$



- Compress by 2 and time-reversal(whichever comes first), $x(-2t+3)$



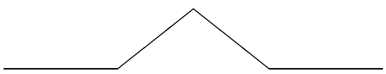
- (Check)

let $x(-2t+3) = x(0)$,	we have $t = 3/2$	indeed $y(3/2) = x(-2t+3) = x(0)$
let $x(-2t+3) = x(-1)$,	we have $t = 2$	indeed $y(2) = x(-2t+3) = x(-1)$
let $x(-2t+3) = x(2)$,	we have $t = 1/2$	indeed $y(1/2) = x(-2t+3) = x(2)$

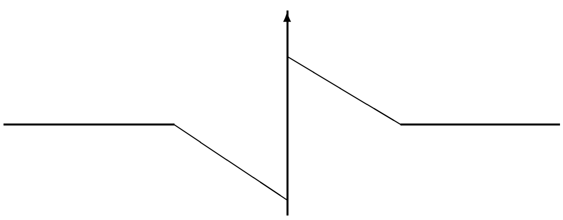
4 Signal characteristics

1. Even/Odd

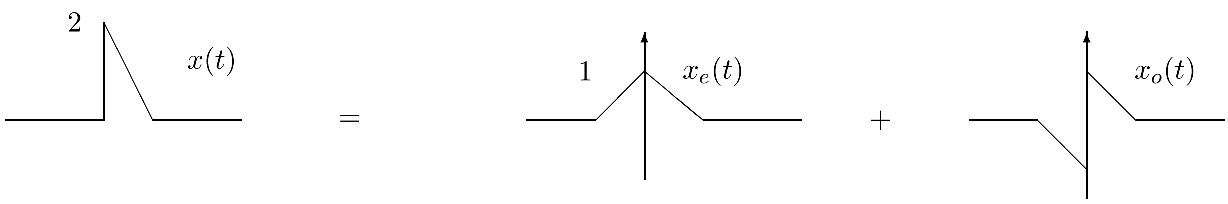
   - Even function(signal): $x(t) = x(-t)$, $\forall t$, symmetric about the origin,



   - Odd function(signal): $x(t) = -x(-t)$, $\forall t$, anti-symmetric about the origin,



   - (Fact) Every signal $x(t)$ can be decomposed as a sum of even and odd signals: $x(t) = x_e(t) + x_o(t)$, where $x_e(t) \equiv \mathcal{E}v\{x(t\} \equiv \frac{x(t)+x(-t)}{2}$ and $x_o(t) \equiv \mathcal{O}d\{x(t\} \equiv \frac{x(t)-x(-t)}{2}$.



2. Periodic signals: $x(t) = x(t+nT)$, $\forall t$. T is its period.

   Sum of several continuous-time periodic signals may not be period, depending on the relationship among their fundamental periods.

3. Causal signal: $x(t) = 0, \forall t < 0$; anticausal signal: $x(t) = 0, \forall t > 0$.

5 Basic Signal Models

1. Complex exponential signals:

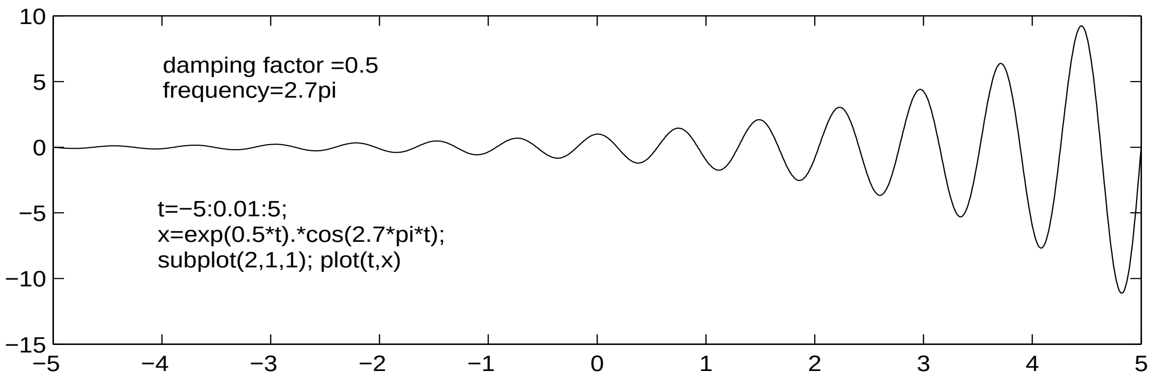
$$\begin{aligned} x(t) &= Ce^{st} \equiv Ae^{j\phi}e^{(\sigma+j\omega)t} \cdots\cdots s \equiv \sigma + j\omega \text{ in Laplace transform} \\ x[n] &= Cz^n \equiv Ae^{j\phi}(re^{j\omega})^n \ \cdots\cdots z \equiv re^{j\omega} \quad \text{in Z transform} \end{aligned}$$

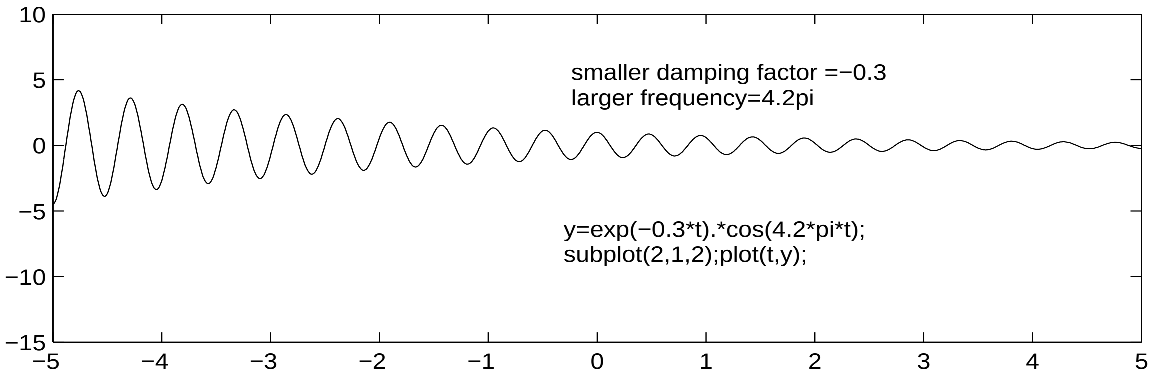
   (a) $\omega = 0$ and $\phi = 0$, $x(t) = Ae^{\sigma t}$ is real exponential(non-oscillating),
      i. $\sigma > 0$ or $r > 1$: exponentially growing,
      ii. $\sigma < 0$ or $r < 1$: exponentially decaying,
      iii. $\sigma = 0$ or $r = 0$: constant.

   (b) $\omega \neq 0$,
      i. $\sigma = 0$ or $r = 1$: $x(t) = Ae^{j\phi}e^{j\omega t} = A\cos(\omega t + \phi) + jA\sin(\omega t + \phi)$ or $x[n] = Ae^{j\phi}e^{j\omega n}$ is a non-damping *cisoid*(complex sinusoid). $\cdots\cdots$ will see in Fourier transforms.
      ii. $\sigma > 0$ or $r > 1$: $x(t) = Ae^{\sigma t}e^{j(\omega t+\phi)}$ is an exponentially increasing(unstable) cisoid
      iii. $\sigma < 0$ or $r < 1$: $x(t) = Ae^{\sigma t}e^{j(\omega t+\phi)}$ is an exponentially damped(decreasing) cisoid

   [Euler's formula]

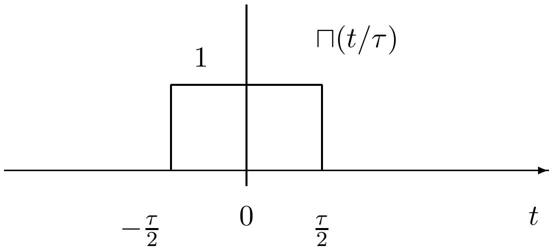
$$\begin{aligned} e^{j\omega t} &\equiv \cos\omega t + j\sin\omega t \quad \text{and} \quad e^{-j\omega t} \equiv \cos\omega t - j\sin\omega t \\ \cos\omega t &= \frac{1}{2}(e^{j\omega t} + e^{-j\omega t}) \quad \text{and} \quad \sin\omega t = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t}) \end{aligned}$$



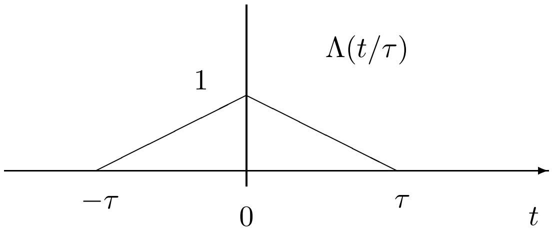


- On the left-hand-side, $\sigma < 0$, $e^{st} \to 0$ as $t \to \infty$, and the magnitude $|\sigma|$ controls the envelope (damping factor); as $|\sigma| \nearrow$ increases, the envelope $\searrow$ decays faster.
- Along the $j\omega$-axis, ω controls the angular frequency of the rotating phasor.

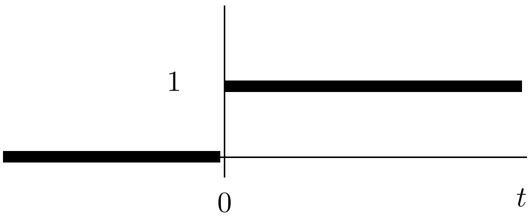
2. Rectangular pulse: $\sqcap(t/\tau)$



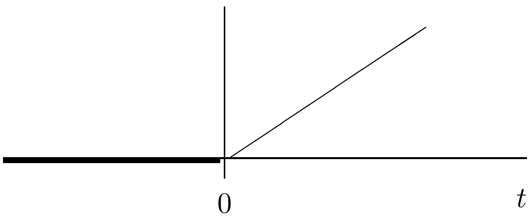
3. Triangular pulse: $\Lambda(t/\tau)$



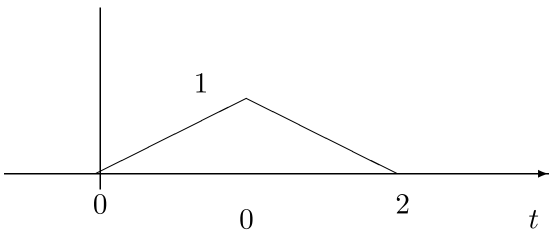
4. Unit step function: $u(t) = \int_{-\infty}^{t} \delta(\lambda) d\lambda = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$



5. Unit ramp function: $r(t) = \int_{-\infty}^{t} u(\lambda) d\lambda = tu(t)$:



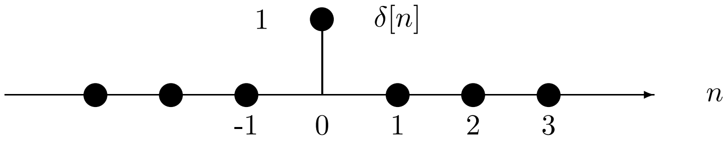
[Ex] $tu(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$



6 Impulse function

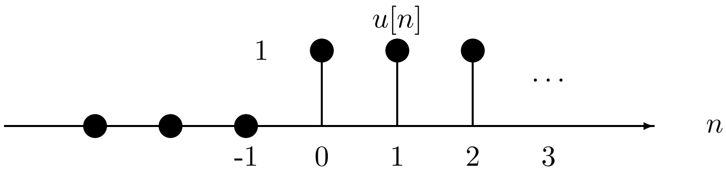
6.1 Unit impulse (Kronecker delta function) in discrete time

$$\delta[n] \equiv \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$



1. unit step function:

$$u[n] \equiv \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$



   - difference equation: $\delta[n] = u[n] - u[n-1]$
   - running sum: $u[n] = \sum_{m=-\infty}^{n} \delta[m]$

2. sampling property:

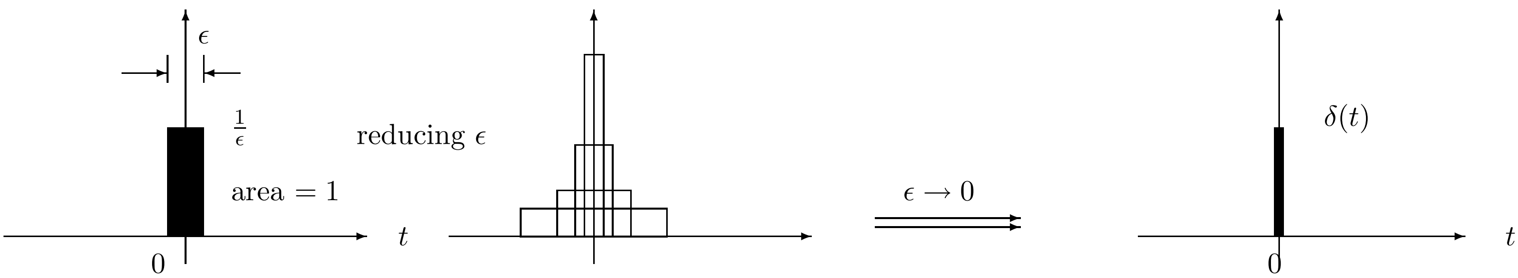
$$\begin{aligned} x[n]\delta[n] &= x[0]\delta[n] \ \text{ similar to differentiation in CT} \\ x[n]\delta[n-n_0] &= x[n_0]\delta[n-n_0] \ \text{ similar to integration in CT} \end{aligned}$$

3. sifting property:

$$\begin{aligned} x[n] &= \sum_k x[k]\delta[n-k] \cdots\cdots \text{ convolution of } x[n] \text{ and } \delta[n] \\ &= \cdots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \cdots \\ &\quad \text{any sequence can be expressed as a sum of scaled/shifted impulses .} \end{aligned}$$

6.2 Unit impulse function(Dirac delta function): $\delta(t)$, in continuous time

$$\delta(t) = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \sqcap (\frac{t}{\epsilon}) \ = \ \frac{d}{dt}u(t), \ \text{ a pulse with unit area and zero width, located at } t = 0$$



Properties of $\delta(t)$:

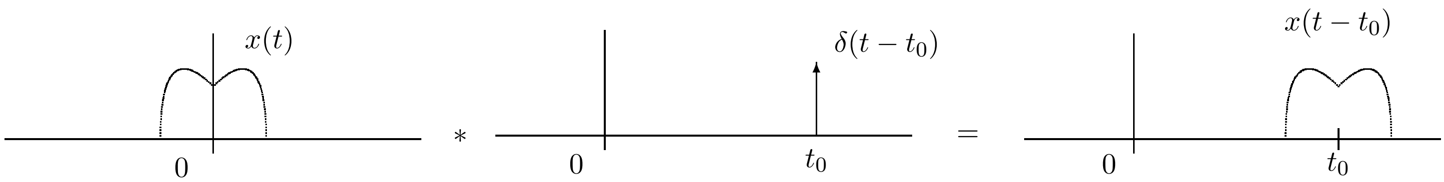
1. **sifting**: $\boxed{\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)dt = x(t_0)}$

$$\begin{aligned}\int x(t)\,\delta(t-t_0)dt &= \int x(t_0)\,\delta(t-t_0)dt \\ &= x(t_0)\int \delta(t-t_0)dt \\ &= x(t_0)\end{aligned}$$

Example: $\int_{-\infty}^{\infty} e^{-2t}\delta(t-3)dt = e^{-6}$

2. $\boxed{\text{Convolution of } x(t) \text{ with } \delta(t-t_0) \text{ is } x(t-t_0).}$

which is useful in proving the sampling theorem and modulation/demodulation.



Pf. $x(t) * \delta(t-t_0) = \int x(\tau)\delta(t-t_0-\tau)d\tau = \int x(\tau)\delta(\tau-t+t_0)d\tau = x(t-t_0)$.

Question: What is $x(t) * [\delta(t-t_0)+\delta(t+t_0)] * [\delta(t-t_0)+\delta(t+t_0)]$? (demodulation)

3. Other properties:

- $x(t)\delta(t) = x(0)\delta(t)$.
- $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$.
- $\delta(t) = \frac{d}{dt}u(t)$.
- $\int_{-\infty}^{t} \delta(\tau)d\tau = u(t)$.
- $\delta(at) = \frac{1}{|a|}\delta(t)$.
- $\delta(-t) = \delta(t)$.
- $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$.
- $\int_{-\infty}^{\infty} \delta(\tau)\delta(t-\tau)d\tau = \delta(t)$.
- $\int_{t=0}^{4} 3\delta(t-2)dt = 3$.
- $\int_{t=2}^{4} \delta(t-6)dt = 0$.

7 Systems

process an input $x(t)$ to produce an output $y(t)$: cause $\rightarrow$ $\boxed{\text{system}}$ $\rightarrow$ effect

1. Physical models: an electric circuit

2. Block diagrams

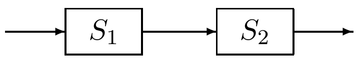
   - continuous-time: using integrators, adders, and gains (and multipliers)
   - discrete-time: using delays, adders, and gains (and multipliers)

3. Mathematical system equations:

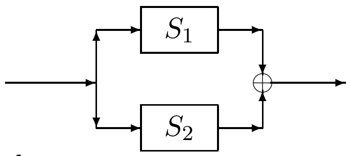
   - integro-differential equation: $y(t) = \frac{dx(t)}{dt} \int_{-\infty}^{t} [4x(\tau) - y(\tau)]d\tau$.
   - difference equation: $y[n] = ay[n-1] + bx[n]$

4. Interconnection of systems:

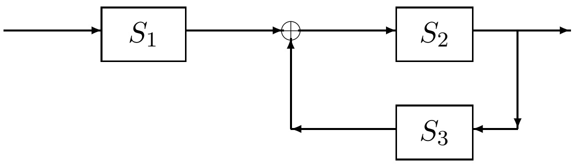
   (a) series(cascade):

   (b) parallel:

   (c) feedback:

8 Classification of Systems

1. Continuous-time or Discrete-time,

2. Analog or Digital,

3. Memoryless: the o/p of a memoryless system at time t_0 depends only on its input at the same time instant t_0. Memory is associated with storage of energy in physical systems and storage registers in digital computers.

   Delay, capacitor, and inductor elements are not allowed for a memoryless system. Thus, a resistive voltage divider is a memoryless system, while an RC lowpass circuit has memory.

   [Ex] $y[n] = x^2[n] + x[n], y(t) = 10x(t)$: memoryless.
   A voltage divider: $v_o(t) = \frac{R_2}{R_1+R_2} v_i(t)$ is also memoryless

   [Ex] $y(t) = \int_{-\infty}^{t} x(\tau)d\tau, y[n] = x[n-2], y[n] = (x[n-1] + x[n] + x[n-1])/3$: with memory.
   An electric example is a current source $i(t)$ connected to a capacitor, then the voltage across the capacitor is $v(t) = \frac{1}{C} \int_{-\infty}^{t} i(\tau)d\tau$.

4. Invertible: A system is invertible if the cascade of this system with its *inverse system* yields an output which is the input to the first system.

$$x[n] \rightarrow \boxed{\text{system}} \rightarrow y[n] \rightarrow \boxed{\text{inverse system}} \rightarrow w[n] \equiv x[n]$$

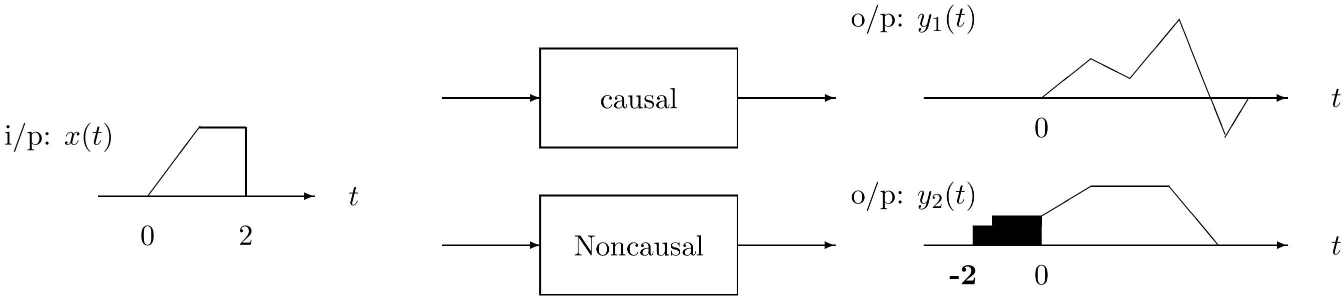
   Examples:

   (a) System: $y(t) = 2x(3t)$; its inverse system: $w(t) = \frac{1}{2}y(\frac{t}{3})$

   (b) A running-sum system: $y[n] = \sum_{k=-\infty}^{n} x[k]$; its inverse system satisfies a difference equation: $w[n] = y[n] - y[n-1]$.

   Inverse processing is important in equalization, lossless coding, *etc.*

5. **Causal**(nonanticipative) and noncausal: A causal system's output $y(t_0)$ can only depends on the present and past inputs: $\{x(\tau), \tau \leq t_0\}$, *i.e.,* the system cannot anticipate the future input.



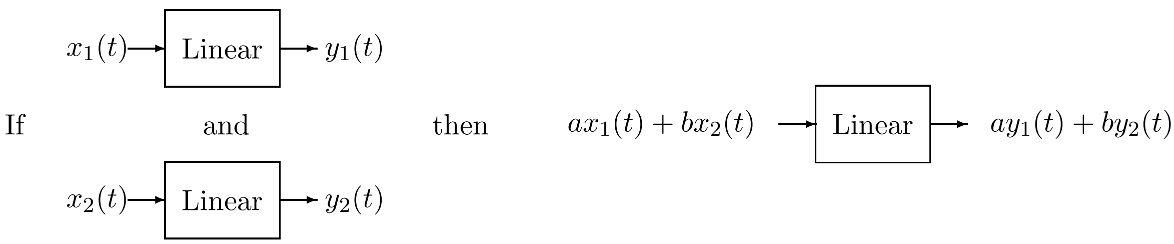
   Later, we will show that, for a causal LTI system, its impulse response $h(t) = 0, \ \forall t < 0$.

   [Ex] $y[n] = x[n] - x[\underline{n+1}]$, and $y(t) = x(\underline{t+1})$ are noncausal, because the output $y(2)$ at present time $t = 2$ depends on the future input $x(3)$ at $t = 3$.

   - A memoryless system is also causal.
   - All physical systems with time as the independent variable are causal. There are practical systems that are not causal:
     - Physical systems for which time is not the independent variable *e.g.,* the independent variable is (x, y) as in an image. $y[n] = (x[n-1] + x[n] + x[n+1])/3$, take the average of three neighboring data.
     - Processing of signals is not in real time, *e.g.,* the signal has been recorded or generated in a computer.

6. Stable: A system is stable in the sense of **bounded-input, bounded-output**(BIBO) if the output $y(t)$ is bounded for a bounded input $x(t)$, i.e., if $|x(t)| \leq B_1, \forall t$, then $\exists B_2 < \infty$, $\ni$ $|y(t)| \leq B_2, \forall t$. For a stable linear-time-invariant system, its impulse response must be absolutely integrable: $\int_{-\infty}^{\infty} |h(t)|dt < \infty$. (to be shown later)

7. **LINEAR**: linear combination of inputs lead to linear combination of corresponding outputs (superposition):



Suppose $y_1(t) = \mathcal{H}[x_1(t)]$, and $y_2(t) = \mathcal{H}[x_2(t)]$. If the output due to the input $a_1x_1(t) + a_2x_2(t)$ is equal to $a_1y_1(t) + a_2y_2(t)$ then the system $\mathcal{H}$ is linear, *i.e.*, check if

$$\mathcal{H}[a_1x_1(t) + a_2x_2(t)] \stackrel{?}{=} a_1y_1(t) + a_2y_2(t)$$

- Linearity does not allow $x^2(t)$, $|x(t)|$.

8. **TIME-INVARIANT**(TI): a delayed input leads to a corresponding delayed output:

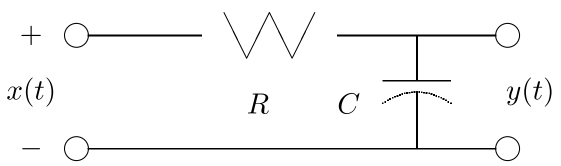
$$\text{If} \quad x(t) \longrightarrow \boxed{\begin{array}{c}\text{Time}\\ \text{Invariant}\end{array}} \longrightarrow y(t) \quad \text{then} \quad x(t-\tau) \longrightarrow \boxed{\begin{array}{c}\text{Time}\\ \text{Invariant}\end{array}} \longrightarrow y(t-\tau)$$

for all $x(t)$ and τ.

We need to verify if the following is true:

$$\mathcal{H}[x(t-\tau)] \stackrel{?}{=} \{\mathcal{H}[x(t)]\}|_{t:t-\tau} = y(t-\tau), \quad \forall\tau$$

(a) A system is time-invariant if its system parameters are fixed over time.



An RC lowpass filter with constant resistance R and C is time-invariant: $RC\frac{d}{dt}y(t) + y(t) = x(t)$. If the resistance $R(t)$, changes with time, then the system becomes time-varying: $R(t)C\frac{d}{dt}y(t) + y(t) = x(t)$, because its system parameters are not constant over time.

(b) Time-invariance does not allow $x(2t)$, $x(-t)$, $tx(t)$.

(c) Ex. A compressor with $y[n] = x[Mn]$ is not TI because $y[n-n_0] = x[M(n-n_0)] \neq x[Mn-n_0]$. Be careful in distinguishing $x[M(n-n_0)]$ and $x[Mn-n_0]$.

9 LTI systems

1. Many man-made and naturally occurring systems can be modeled as linear time-invariant (LTI) systems. Ex. resistors(R), inductors(L), capacitors(C).

2. $\boxed{\text{A fixed-coefficient linear differential equation (made up from RLC components) are LTI.}}$ (CT)

3. $\boxed{\text{A fixed-coefficient linear difference equation (delay, gain, adder) are LTI.}}$ (Discrete-Time)

4. (Ex) Is the following system linear-time-invariant?

$$y(t) = x(t)g(t)$$

where $x(t)$ and $y(t)$ denote the input and output, respectively.

- Linear, yes. Let $y_1(t) \equiv x_1(t)g(t)$ and $y_2(t) \equiv x_2(t)g(t)$. Suppose

$$x(t) = a_1x_1(t) + a_2x_2(t)$$

then

$$\begin{aligned}\mathcal{H}[x(t)] &= \underline{x(t)g(t)} \\ &= a_1x_1(t)g(t) + a_2x_2(t)g(t) \\ &= \underline{a_1y_1(t) + a_2y_2(t)}\end{aligned}$$

- Time-varying. Let $y(t) = x(t)g(t)$. Suppose $x_d(t) = x(t-\tau)$.

$$\begin{aligned}\mathcal{H}[x_d(t)] &= x(t-\tau)g(t) \\ &\neq y(t-\tau) = x(t-\tau)g(t-\tau)\end{aligned}$$

- If $g(t) = \cos\omega_c t$, we call it *linear modulation.*

9.1 Examples

1. $\boxed{\frac{d}{dt}y(t) + 10y(t) = x(t) \text{ is a causal LTI system.}}$ $y[n] = 0.9y[n-1] + x[n]$ is causal LTI, too.

2. In general, $\frac{d^n}{dt^n}y(t) + a_{n-1}\frac{d^{n-1}}{dt^{n-1}}y(t) + \cdots + a_0 y(t) = b_m \frac{d^m}{dt^m}x(t) + \cdots + b_1\frac{d}{dt}x(t) + b_0 x(t)$ is a linear system.

3. $\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \text{ is an LTI system.}}$ reads as "$y(t)$ is the convolution of $x(t)$ and $h(t)$."

   [Pf] The proof for linearity is omitted.

   Let $\hat{x}(t) = x(t-\lambda)$, then

$$\begin{aligned}\hat{y}(t) &: \text{ is the response to the input } \hat{x}(t) \\ &= \int_\tau \hat{x}(\tau)h(t-\tau)d\tau \\ &= \int_\tau x(\tau-\lambda)h(t-\tau)d\tau \\ &\quad \text{let } s \equiv \tau - \lambda, \text{ sorry for so many dummy variables} \\ &= \int_s x(s)h(t-\lambda-s)ds \\ y(t-\lambda) &= \int_\tau x(\tau)h(t-\lambda-\tau)d\tau, \;\; \text{from the definition of } y(t) = \int x(\tau)h(t-\tau)d\tau\end{aligned}$$

4. (a) $y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$ is LTI, too. (commutative law for convolution)

   (b) $y[n] = \sum_{m:-\infty}^{\infty} x[m]h[n-m] = \sum_{m:-\infty}^{\infty} x[n-m]h[m]$ is LTI. (discrete-time convolution)

   (c) $\underline{y(t) = \int_{-\infty}^{t} x(\tau)h(t-\tau)d\tau}$ is a <u>causal LTI</u> system.

   $y[n] = \sum_{m:-\infty}^{n} x[m]h[n-m]$ is a causal LTI system.

   (d) $y(t) = \int_{-\infty}^{t} x(t-\tau)h(\tau)d\tau$ is a noncausal, linear, time-varying system.

   [Pf] Say $h(t) = 1$, then $y(t) = \int_{-\infty}^{t} x(t-\tau)d\tau$ we can see that $y(0) = \int_{-\infty}^{0} x(-\tau)d\tau = \int_0^\infty x(u)du$, which means that $y(0)$ is the integral of future input $x(0+)$ upto $x(\infty)$. Noncausal!

5. (a) $\frac{d}{dt}y(t) + ty(t) = x(t)$ is a linear system.

$$\frac{d}{dt}y_1(t) + ty_1(t) = x_1(t) \quad \frac{d}{dt}y_2(t) + ty_2(t) = x_2(t)$$

$$\alpha_1\frac{d}{dt}y_1(t) + \alpha_2\frac{d}{dt}y_2(t) + t[\alpha_1 y_1(t) + \alpha_2 y_2(t)] = \alpha_1 x_1(t) + \alpha_2 x_2(t)$$

$$\frac{d}{dt}[\alpha_1 y_1(t) + \alpha_2 y_2(t)] + t[\alpha_1 y_1(t) + \alpha_2 y_2(t)] = \alpha_1 x_1(t) + \alpha_2 x_2(t)$$

   Thus the response to the input $\alpha_1 x_1(t) + \alpha_2 x_2(t)$ is $\alpha_1 y_1(t) + \alpha_2 y_2(t)$.

   (b) $\frac{d}{dt}y(t) + 10y(t) + 5 = x(t)$ is a nonlinear system.

(c) $y(t) = x(t^2)$ is linear, noncausal, and time-varying.
$y(2)$ depends on $x(4)$; thus noncausal. Let $\hat{x}(t) = x(t-\tau)$, then $\hat{y}(t) = \hat{x}(t^2) = x(t^2-\tau) \neq y(t-\tau) = x((t-\tau)^2)$; thus time-varying.

(d) $y(t) = \sin[x(t)]$ is time-invariant.

(e) $y(t) = x(2t)$ is NOT time-invariant.
[Pf] Suppose $x(t) \Longrightarrow y(t) = x(2t)$. Let $\hat{x}(t) = x(t-\tau)$, then the output from $\hat{x}(t)$ is

$$\hat{y}(t) \;=\; \hat{x}(2t) = \hat{x}(s)|_{s=2t} = x(s-\tau)|_{s=2t} = x(2t-\tau)$$

Compared with the delayed output $y(t-\tau)$:

$$y(t-\tau) \;=\; y(s)|_{s=t-\tau} = x(2s)|_{s=t-\tau} = x(2(t-\tau)) = x(2t-2\tau)$$

We find that $\hat{y}(t) \neq y(t-\tau)$, it is NOT TI!

(f) $ny[n] + a_1 y[n-1] = x[n]$ is a linear and time-varying(because of n before $y[n]$).

(g) $y[n] + 0.8y[n-1] + 5 = x[n]$ is a nonlinear system(because of the constant 5).

(h) $y[n] = x[n^2]$ is linear, noncausal, and time-varying.

(i) $y[n] = sin(x[n])$ is time-invariant.

9.2 Questions

1. If some system: $x(t) \longrightarrow y(t)$ is linear, and another system: $y(t) \longrightarrow z(t)$ is linear, too, Is the cascaded system $x(t) \longrightarrow z(t)$ linear?

2. If some system: $x(t) \longrightarrow y(t)$ is linear, and another system: $x(t) \longrightarrow z(t)$ is linear, too, Is the parallel system $x(t) \longrightarrow w(t) = y(t) + z(t)$ linear?

3. Redo the above two questions, when the systems are time-invariant.

4. Is the cascaded connection of two nonlinear systems is nonlinear?

5. (Oppenheim *et al.*) Consider three systems with the following input/output relationships:

$$\begin{aligned} \text{system 1} &: \; y[n] = \begin{cases} x[n/2], & n: \text{ even} \\ 0, & n: \text{ odd} \end{cases} \\ \text{system 2} &: \; y[n] = x[n] + \frac{1}{2}x[n-1] + \frac{1}{4}x[n-2] \\ \text{system 3} &: \; y[n] = x[2n] \end{aligned}$$

Suppose these systems are cascaded in series. Find the input/ouput relationship for the overall cascaded system. Is it linear? Is it time-invariant?

6. Can you generalize the definitions of linearity and time-invariance for a system processing 3D signals with 2 independent variables: $\underline{s}(t_1, t_2)$, where $\underline{s} = [s_1, s_2, s_3]$?